

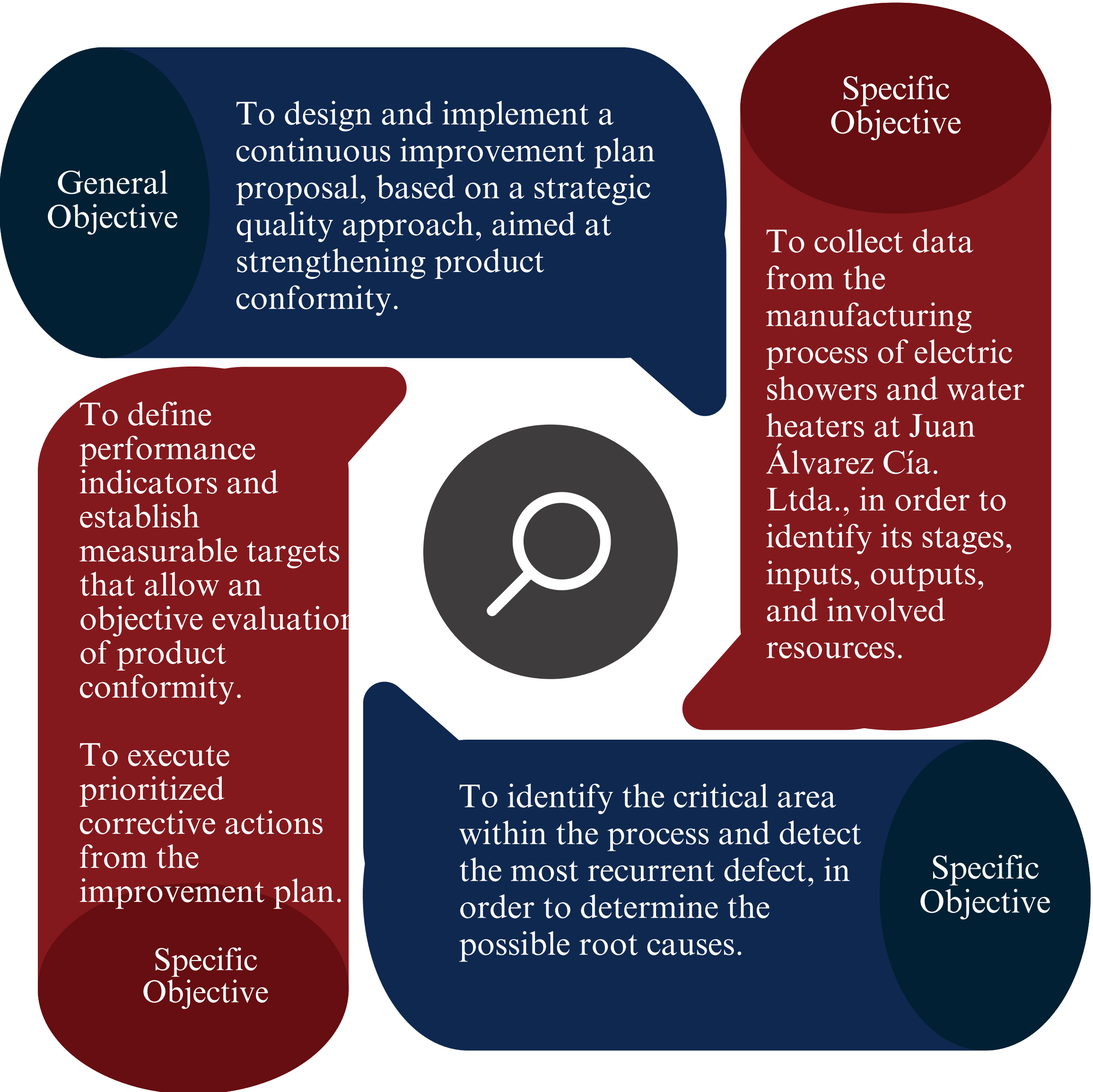
Strategic Quality Approach for Product Conformity: Electric Shower and Water Heater Line of Juan Álvarez Cía. Ltda.

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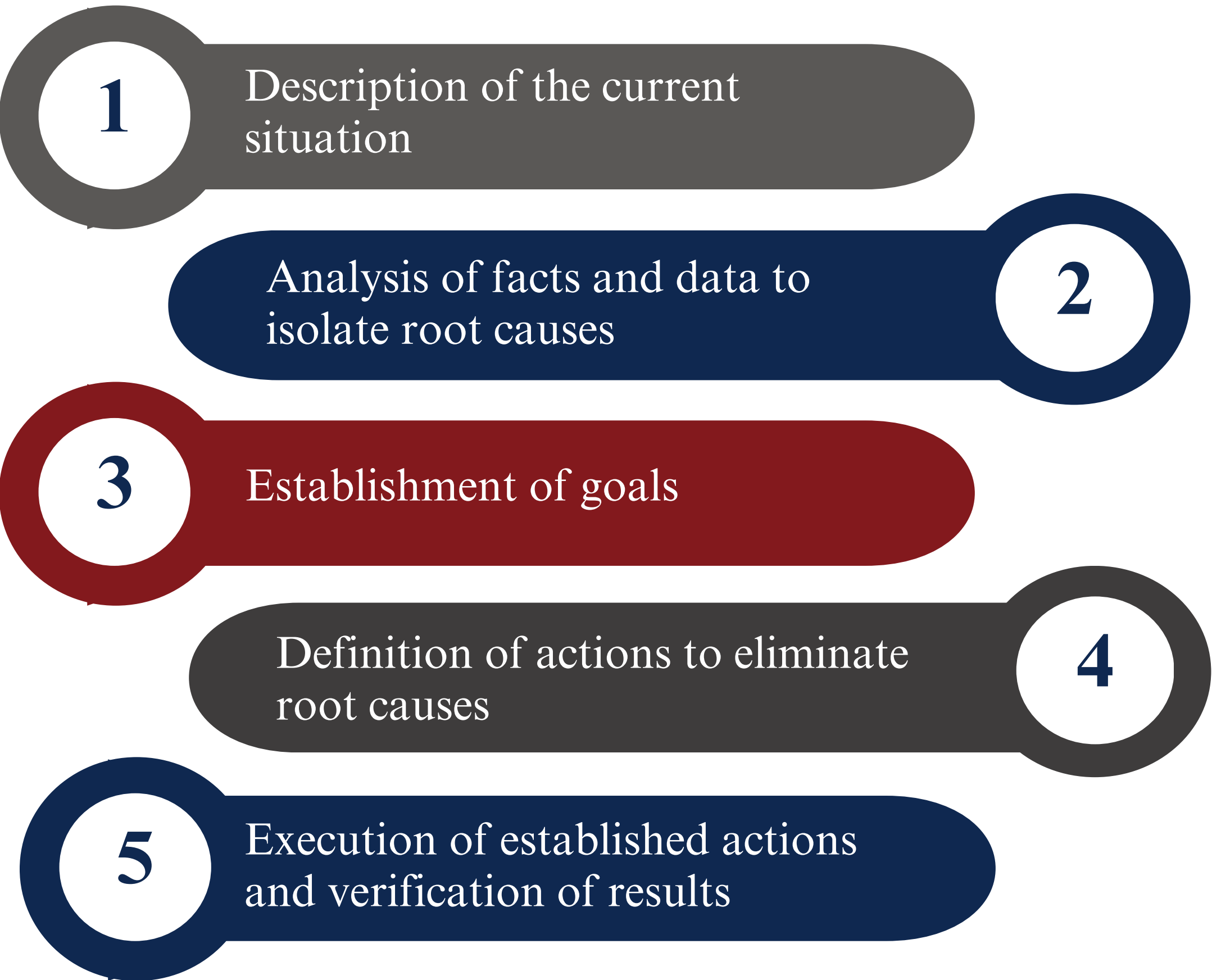
Abstract

A strategic approach was implemented to control production variability. Operational formalization generated a 29.7% improvement in product conformity, which was the determining factor in ensuring quality.

Objectives



Methodology



Curricular Integration

Quality Control: Statistical foundations, Basic Quality Tools.
Process Engineering and Ergonomics: Process Diagramming (DOP), operational standardization.

Applied Engineering

- ISO 9001:2015 - Quality Management.
- ISO 10013:2021 - Guidelines for documented information.
- INEN-ISO/IEC 60335-2-35:2012 + Amd.1:2016 Standard.

Keywords

- Product conformity
- Quality Route
- Process standardization
- Plastic injection molding

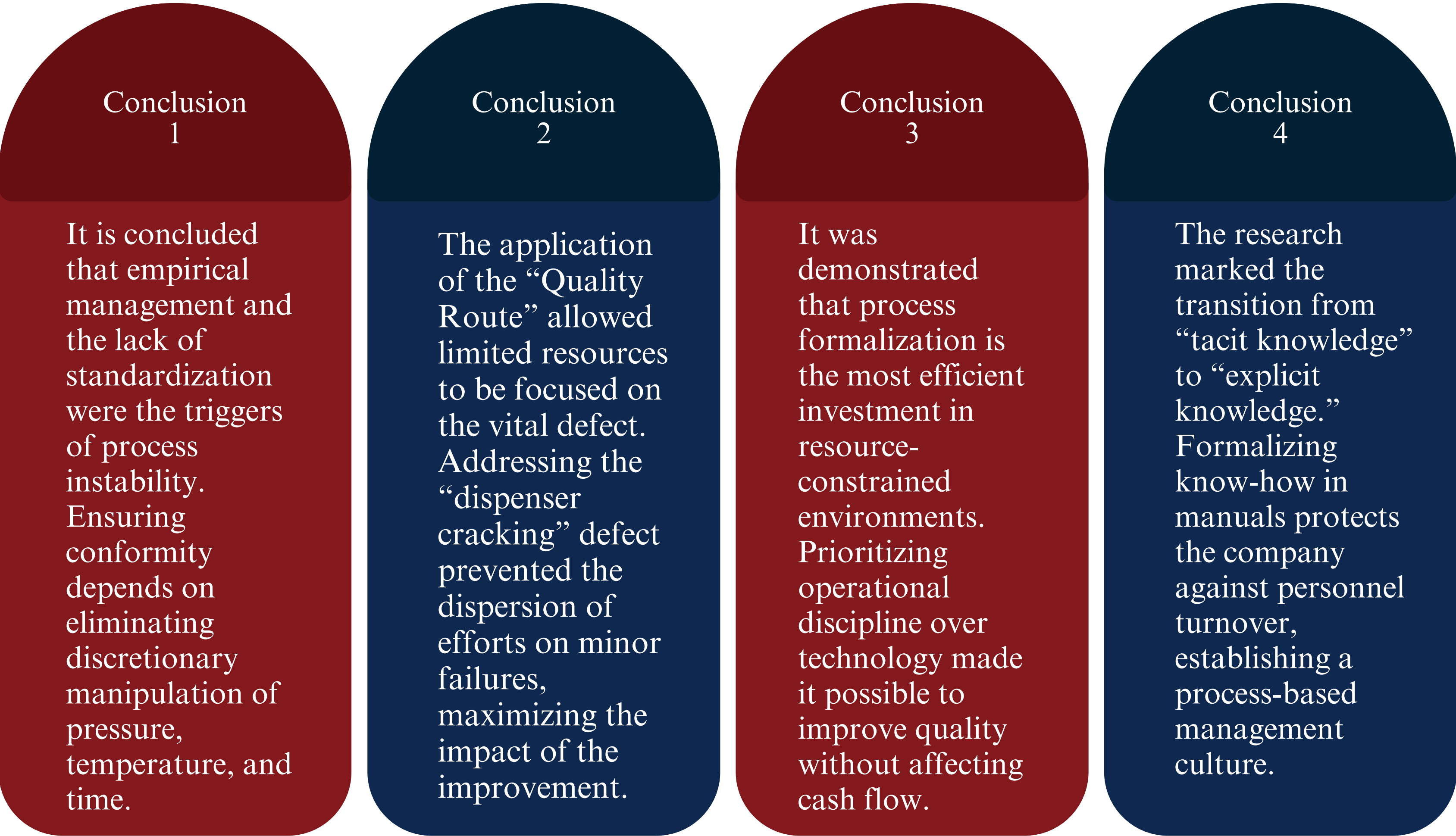
Results

After the intervention, the defect rate decreased from 2.76% to 1.94%. This reduction represents an immediate relative improvement of 29.7%, validating that operational standardization is eliminating product variability.

The financial feasibility of the proposal was demonstrated. Without major investments, the recovered efficiency will prevent the production of 690 defective units per year, optimizing operating costs and freeing productive capacity.

These results represent only the beginning. As operators gain experience and maintain the established order, quality improvement will cease to be temporary and will become permanent and sustainable over time.

Conclusions



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References

- (ITESM, 1996)
- (Gutiérrez & de la Vara, 2013)
- (Zala et al., 2024)

